



ultium[®] EMG

Quick Start Guide

On behalf of Noraxon...

Welcome and congratulations on acquiring your new Ultium EMG System!

This guide will provide you with step-by-step instructions on how to install your new hardware and software, adjust device settings, and record your first data set.

Let's begin by walking through how to install your new hardware.

Note: This is not meant to be a complete user manual, but a guide to help you get started with your system. Refer to the Ultium EMG User Manual (#P-8808) for more detailed operating instructions.

1 System Unboxing

The Ultium EMG System is packed within a reinforced padded box for storage and protection during transport. Upon arrival, carefully remove all contents and verify the following components are present.



Figure 1 – Ultium Receiver (#880)



Figure 2 – Ultium or Core Sensor Charger (#873 or #873C)



Figure 3 – Ultium EMG or Core EMG Sensor (#810 or #816)



Figure 4 – Ultium EMG SmartLead (#842)



Figure 5 – A to B USB Cable (#CBL2)



Figure 6 – Sensor Charger Power Source (#PSU1)



Figure 7 – Sensor Charger to Receiver Cable (#CBL34)

Additional items that may be included with your Ultium EMG System are:

- Double-Sided Sensor Tape (#874E)
- Dual Solid Gel EMG Electrodes (#271S)
- Ultium Travel Case (#CASEU)
- Ultium EMG User Manual (#P-8808)

2 Installing the Hardware

Step 1

Insert the USB cable (#CBL2) into the USB connector on the back of the Ultium Receiver (#880).

Insert the opposite end of the USB cable into an available USB port on the computer.



Step 2

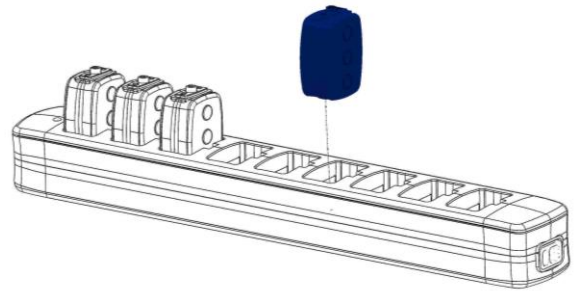
Insert one EMG SmartLead (#842) into each EMG sensor (#810).



Step 3

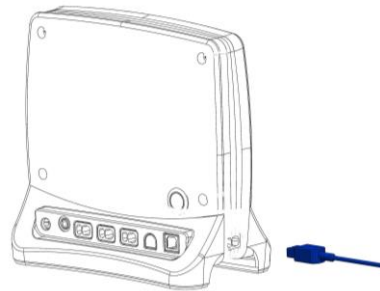
Insert the EMG sensors into the Sensor Charger (#873).

Note: Sensor power button & LED must face the front of the Charger, i.e., toward the charger LEDs.



Step 3

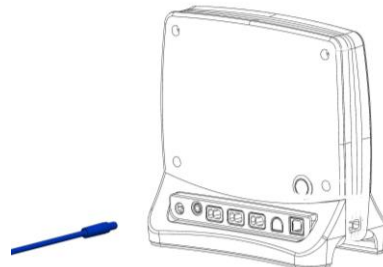
Connect the Receiver to the Sensor Charger using the ports on either side of the Receiver.



Step 4

Insert the power supply (#PSU1) barrel connector into the associated receptacle on the back of the Ultium Receiver to charge sensors.

Note: Sensor LED will show solid orange while charging and turn off when fully charged.



3 Installing the Companion Software - myoRESEARCH™

To utilize the full functionality of the Ultium EMG system, and ensure the system has updated drivers, Noraxon's myoResearch (MR) needs to be installed on the computer.

Note: The Ultium Receiver requires the Noraxon USB device driver which is pre-installed by the MR software installation. It is also available in the [Downloads](#) section of the Noraxon website.

3.1 Software Installation

Within the Ultium Motion System shipment, there is a USB flash drive containing the latest myoRESEARCH software.

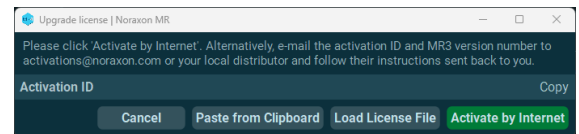
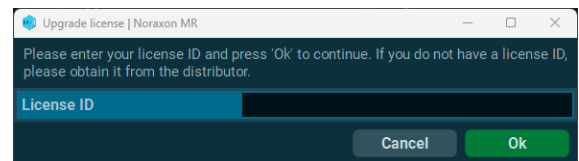
1. Insert the MR USB flash drive into the PC
2. Open the installer application (e.g., noraxon.mr.4.0.0) and follow the instructions
3. After installation, an icon will be created on the desktop



3.2 Companion Software Activation

The companion software must be activated before unrestricted use is possible.

1. Open myoRESEARCH
2. A dialog box will indicate how many more times MR can be opened in trial mode
3. Click **Activate**
4. Enter the License ID provided on your USB flash drive and click **Ok**
5. If you have an internet connection, click **Activate by Internet** for immediate activation
 - a. Alternatively, email the provided activation ID to activation@noraxon.com
 - b. Noraxon Support will email or respond by phone with the Activation Code
 - c. Enter the provided Activation Code to remove any restrictions on use

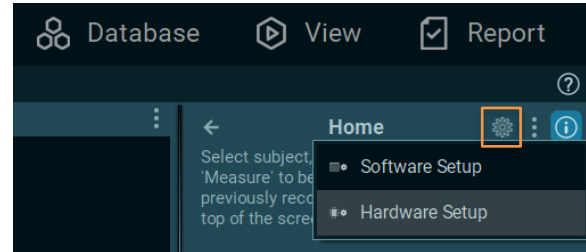


4 Configuring the Hardware

Before the Ultium system can be used, the device software settings must be configured to recognize the different components that make up the system. Follow the instructions below to update the receiver firmware, sensor firmware, and populate sensors to prepare for data collection.

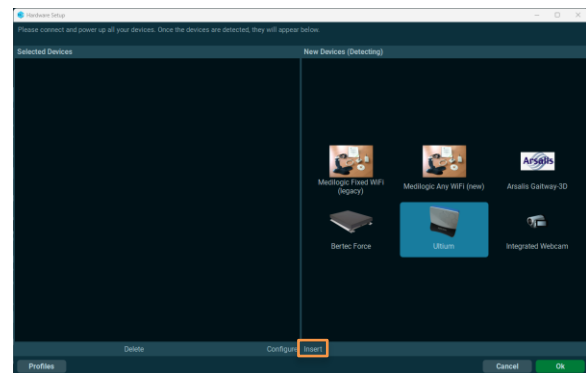
Step 1

Open the MR software, click on the settings icon (⚙️) in the upper right-hand corner, then click **Hardware Setup**.



Step 2

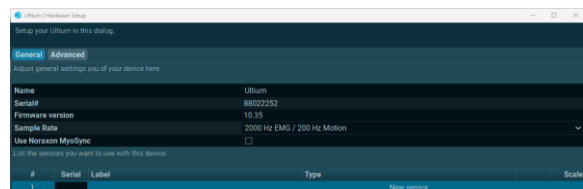
After connecting the Ultium Receiver to a USB port on the computer, select the Ultium Receiver icon in the **New Devices** area, and click **Insert**.



Step 3

The Ultium hardware setup window will appear as shown.

Within the "General" tab, select the desired settings for the system, i.e., name, sample rate, cutoff filters.



Note: If available, firmware updates for the Receiver and Sensors will be indicated here.

If using myoSYNC, click the **Use Noraxon myoSYNC** checkbox.

Additional settings are available within the "Advanced" tab.

Note: If multiple Ultium systems are in use, separate RF network assignments are required for each Receiver.



Step 4

Make sure the EMG or SmartLead Sensors are placed in the Charger(s) and the Charger(s) are connected to the Ultium Receiver, then click **Detect Sensors in Chargers** to load all sensor serial numbers into the MR software.

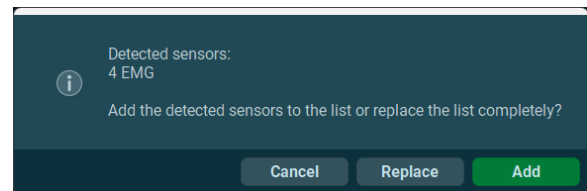
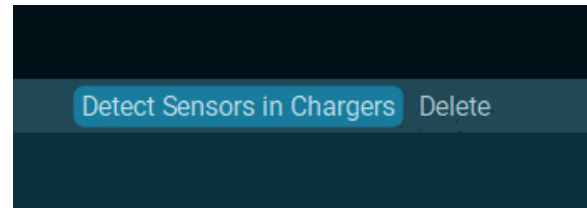
Note: The first serial number in the list is associated with the Sensor closest to the front of the Charger (i.e., toward the Charger LEDs).

Step 5

After detection is complete, click **Replace** to replace all existing sensor serial numbers associated with this device or **Add** to append additional sensor serial numbers to an already existing list.

Step 6

Click **Ok** to save the hardware settings and close the setup window.



List the sensors you want to use with this device.

#	Serial	Label	Type
1	200b5	EMG 1	EMG
2	200ea	EMG 2	EMG
3	200b8	EMG 3	EMG
4	21665	EMG 4	EMG
5			

New sensor



4.1 Configuring Ultium SmartLeads

If you are only using surface EMG leads, continue to Section 5: Recording a Measurement.

Connect the SmartLead to the Ultium sensor. The Ultium sensor LED will flash purple when a SmartLead is connected, and orange when the SmartLead is disconnected.

Click *Detect Sensors in Charger* and the SmartLead will be listed with the SmartLead serial number, not the Ultium sensor serial number.

Note: If the software does not recognize the SmartLead, the sensors may require a firmware update.

Click **OK** to save the hardware settings.

List the sensors you want to use with this device.

#	Serial	Label	Type
1	a00d5	BioMonitor 1	BioMonitor
2	e018e	Insole LT	▼ Insole
3	e018f	Insole RT	▼ Insole
4			

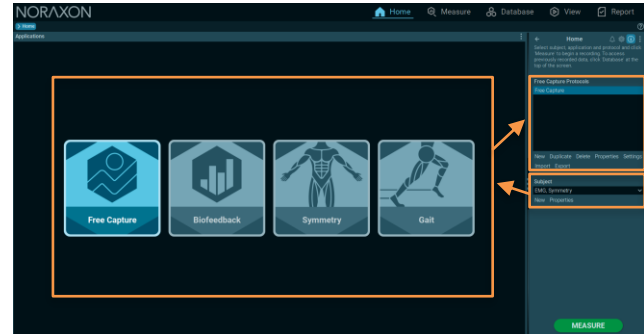
5 Recording a Measurement

Step 1

On the **Home** screen, select or create a subject.

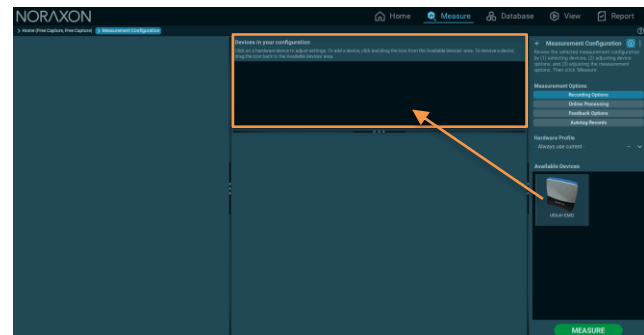
Select an application, then select or create a new protocol.

Click **MEASURE** to proceed.



Step 2

Insert the Ultium EMG device into the configuration by dragging it from the **Available Devices** list to the **Devices in Your Configuration** area.

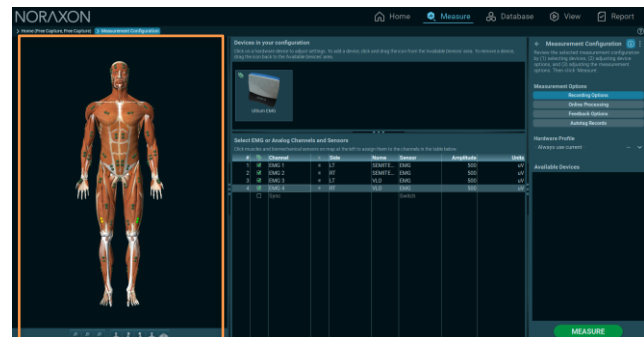


Step 3

Select the channels to include in the measurement by clicking the checkmark.

While a channel is selected, assign a label by selecting the corresponding muscle on the **Muscle Map** or entering the name manually.

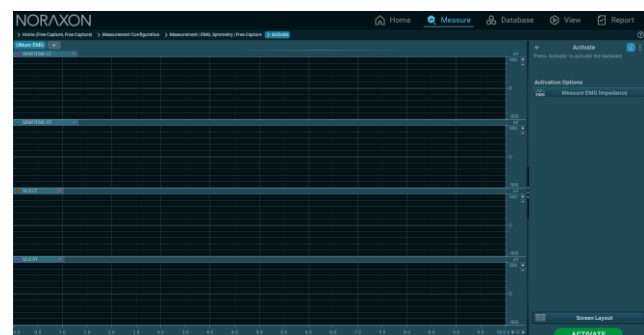
Click **MEASURE** to proceed.



Step 4

Click **ACTIVATE** to begin the measurement.

Note: Additional operations, like measuring EMG impedance or creating an MVC, should be selected before clicking **ACTIVATE**.



Step 5

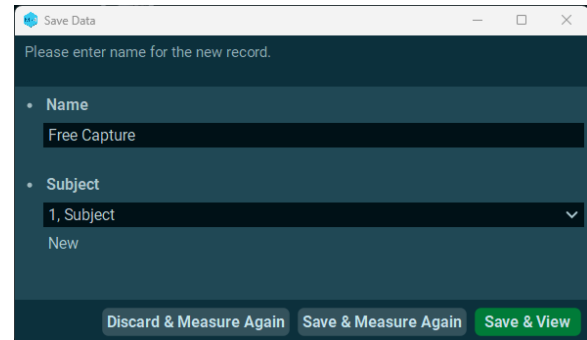
Confirm signal quality and click **RECORD** to begin your recording data.

**Step 5**

Click **STOP** to end the recording, then enter a name for the new record.

Click **Save & View** to view and begin processing the record.

Click **Discard & Measure Again** or **Save & Measure Again** to return to the measurement screen.



6 Viewing a Record

To view a previously recorded measurement, click the **Database** tab. Records are organized by subject name and date measured. Select and click **VIEW** or double-click on the record of interest to open the record in the **View** tab.

7 Further Use Features of MR

There are many additional features built within MR. Such as:

- Multi-activity recordings
- Live biofeedback visual displays
- Online and post-measurement signal processing
- Customized reporting
- Exporting (and importing) of data

To learn more about the features available to you through the system(s) you have purchased, refer to the corresponding Hardware User Manual for this device (#P-8808) and www.noraxon.com.